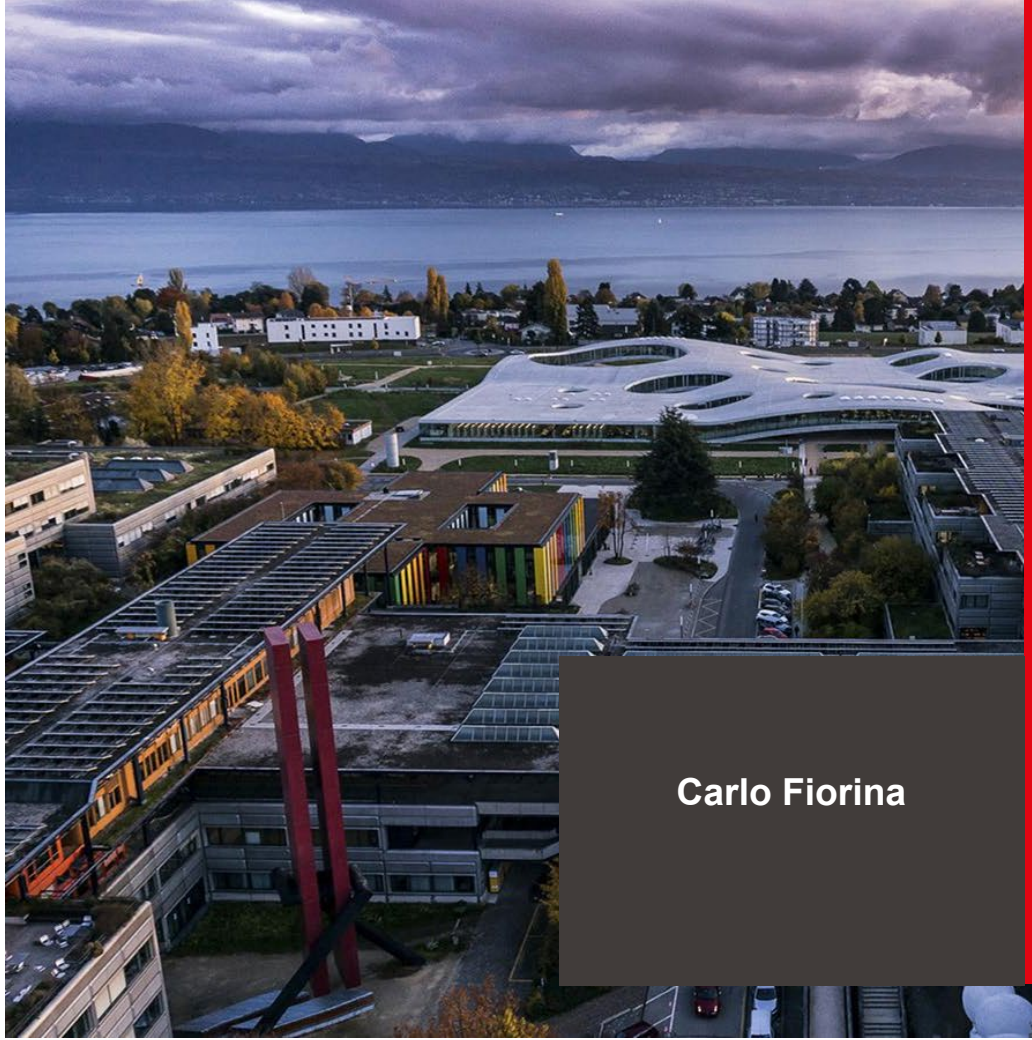


Use of OpenFOAM for multi-physics in nuclear

Carlo Fiorina

École polytechnique fédérale de Lausanne

This workshop

What to expect

- Overview of the modelling capabilities of OpenFOAM
- Lessons learnt
- A crash introduction to OFFBEAT and GeN-Foam

What not to expect

- A full course on the use of OpenFOAM
- A full course on the use of OFFBEAT
- A full course on the use of GeN-Foam

Objectives

- Understanding of modelling capabilities and pros & cons
- How to approach to OpenFOAM-based tools
- References/keywords/best practices for autonomous learning of GeN-Foam and OFFBEAT

This workshop

Format:

- Presentations + interactive discussion
- 1 hands-on exercise

Content:

- Intro to modelling capabilities of OpenFOAM
- Intro to OFFBEAT
- Practical example based on OFFBEAT
- Intro to GeN-Foam

Material

- Slides distributed after workshop, incl. additional material from interactive discussion
- GeN-Foam available on GitLab
- OFFBEAT… Coming soon

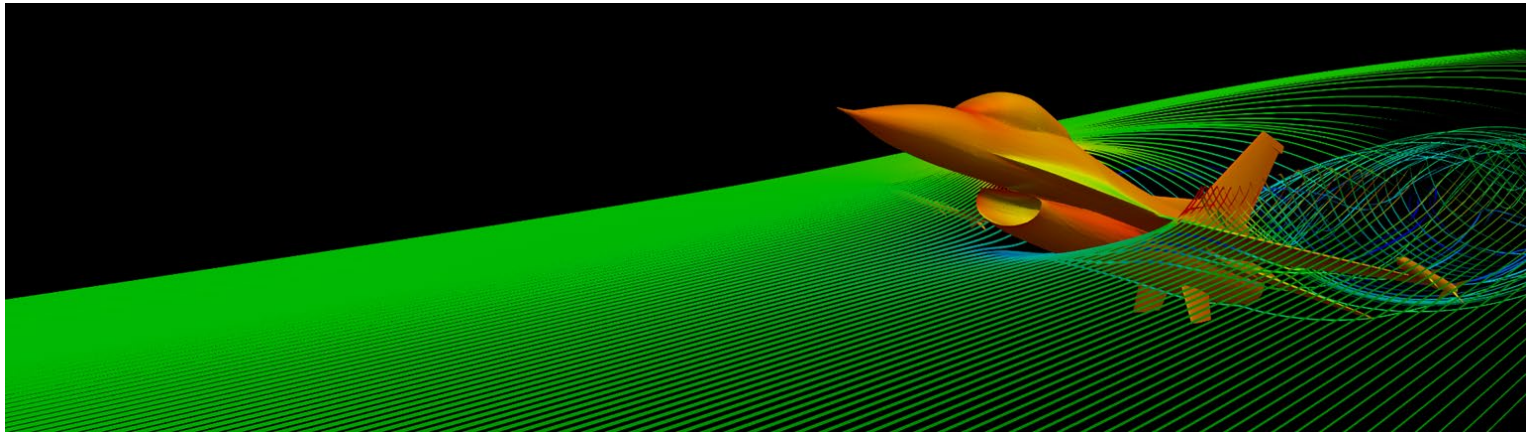
New codes from available numerical libraries

- Fast developments
- Modern algorithms
- Easier maintenance

- What is OpenFOAM?
  - ✓ Distributed as CFD toolbox
  - ✓ ~10k to 20k estimated users worldwide

OpenFOAM

The Open Source CFD Toolbox



OpenFOAM

The Open Source CFD Toolbox

- What is OpenFOAM?
  - ✓ Distributed as CFD toolbox
  - ✓ ~10k to 20k estimated users worldwide
  - ✓ OpenFOAM = Open Field Operation And Manipulation
  - ✓ Essentially a large, well organized, HPC-scalable, C++ library for the finite-volume discretization and solution of PDEs, and including several functionalities like ODE solvers, projection algorithms, and mesh search algorithms
  - ✓ Object-oriented, with a high-level “fail-safe” API

$$\frac{1}{v_i}\frac{\partial \varphi_i}{\partial t} - \Delta(D_i \varphi_i) = S$$

```
fvm::ddt(IV,flux_i])- fvm::laplacian(D,flux_i])= S
```

OpenFOAM: standing on the shoulders of giants

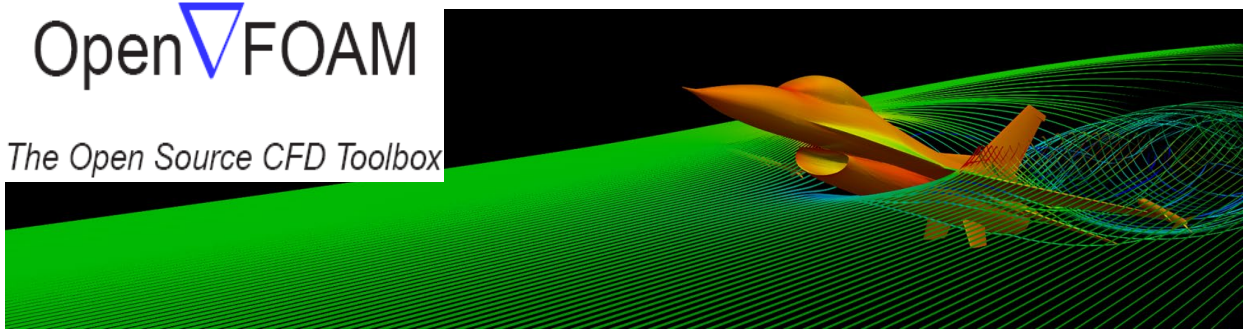
- CFD toolbox
  - Industrial level
- Library (90% of Gen-Foam and OFFBEAT)
  - ✓ Impressively well-written
  - ✓ Object-oriented
  - ✓ Complete (not only discretization and solution)
  - ✓ Quality assured
- Community (20k)
  - ✓ Concurrent developments
  - ✓ Support
- Drawbacks?
  - ✓ Workflow
  - ✓ Documentation

$$\frac{1}{v}\frac{\partial \varphi}{\partial t} = D\Delta\varphi + \left(\frac{\upsilon\Sigma_f * (1.0-\beta_t) * \chi_p}{k_{eff}} - \Sigma_r\right)\varphi + S_d\chi_d$$

```
as
fvMatrix < scalar>
(
fvm::ddt(inverseVelocity,flux)
=fvm::laplacian(D, flux)
+fvm::Sp(nuSigmaF/keff_*(1.0-betaTot)*chiPrompt - sigmaRe-
moval,flux)
+delayedNeutronSource_*chiDelayed
);
```

OpenFOAM

The Open Source CFD Toolbox



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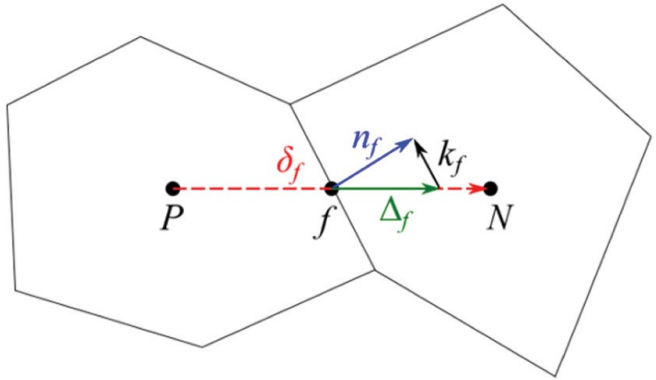
Finite volumes

Conservation of quantities in each cell, with some wise interpolation at faces and techniques to deal with non-orthogonality and skewness

- Pros:
  - ✓ Flexible
  - ✓ Scalable
  - ✓ Conservative
  - ✓ CFD-friendly
  - ✓ Intuitive
- Cons:
  - ✓ Still require familiarity with concepts associated with PDEs (well-posed problems, initial and boundary conditions), geometry creation, meshing, discretization, linear solution, etc.
  - ✓ Require good quality meshes
  - ✓ Max second order in space



Original content

Most of the following is content taken from

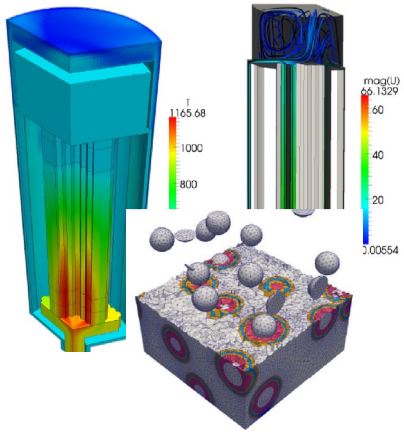
- Carlo Fiorina, Ivor Clifford, Stephan Kelm, Stefano Lorenzi, 2022. “On the development of multi-physics tools for nuclear reactor analysis based on OpenFOAM ®: state of the art, lessons learned and perspectives”. Nuclear Engineering and Design

Use of OpenFOAM for multi-physics

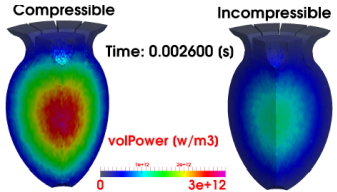
2000-2010
First activities

2010-2015
First widespread use

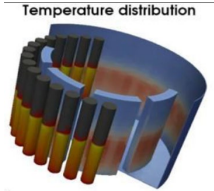
2015-2021
First coordinated and persistent developments



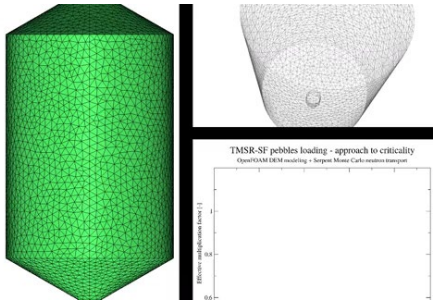
PBMRs and HTRs

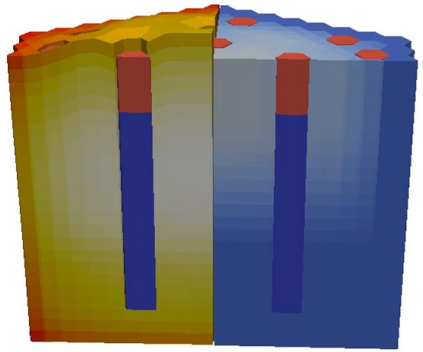


MSRs

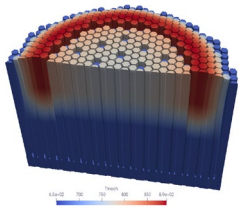


FHRs

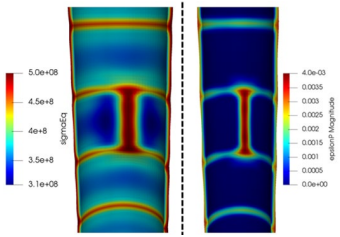




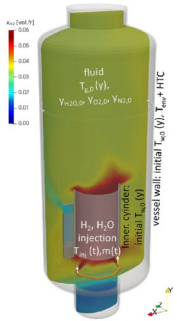
SFRs



GeN-Foam



OFFBEAT



containmentFoam

HTR modelling (PBMR -> I. Clifford)

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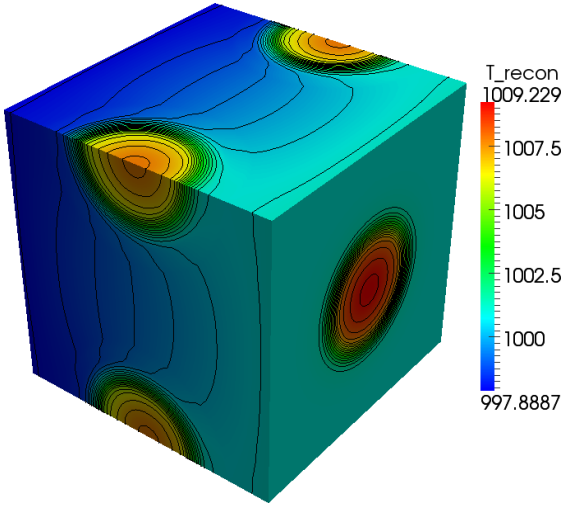
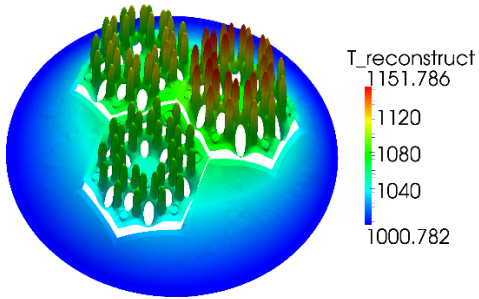
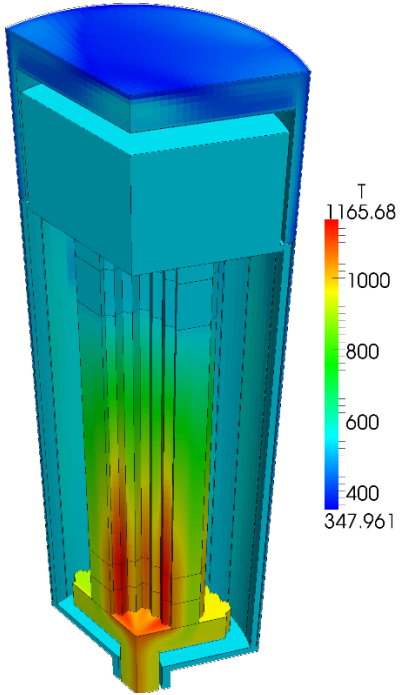
- MHTGR-350MW Benchmark Model



ROM reconstructed solution for TRISO coated particles



ROM reconstructed solution for a fuel element

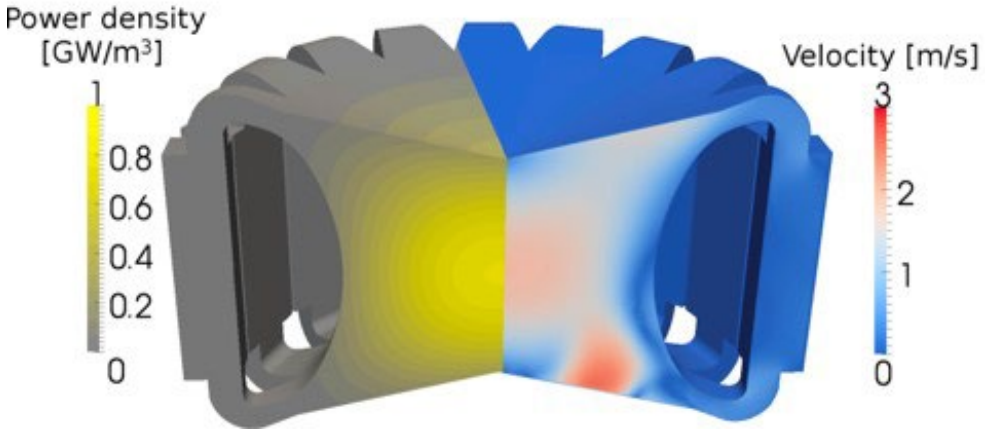


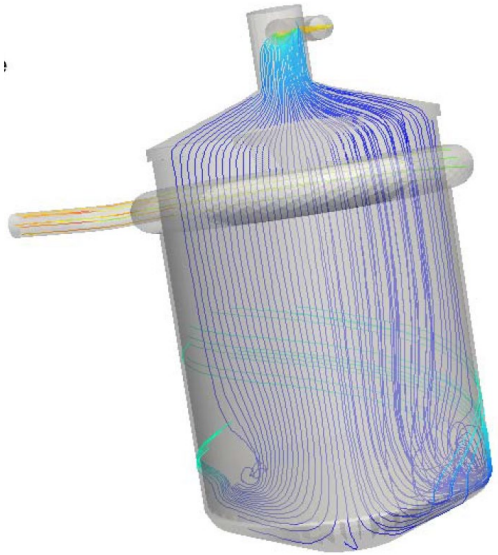
Full-core coarse-mesh thermal-hydraulics

OpenFOAM -> sound TH solvers, ability to access underlying equation terms, customized equation operators for discontinuities, multiple arbitrary meshes, flexible boundary condition design (thermal radiation), classes for POD, time-integration using built-in ODE solvers,…

MSR modelling (M. Aufiero -> PoliMi + CNRS / GeN-Foam)

- MSFR and MSRE





OpenFOAM -> Available CFD solvers, simple implementation of neutron diffusion and precursor transport, arbitrary geometries

MSR modelling: advanced

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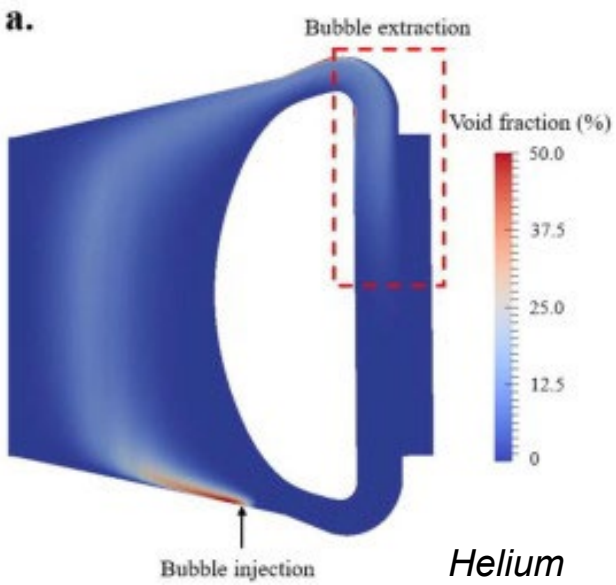
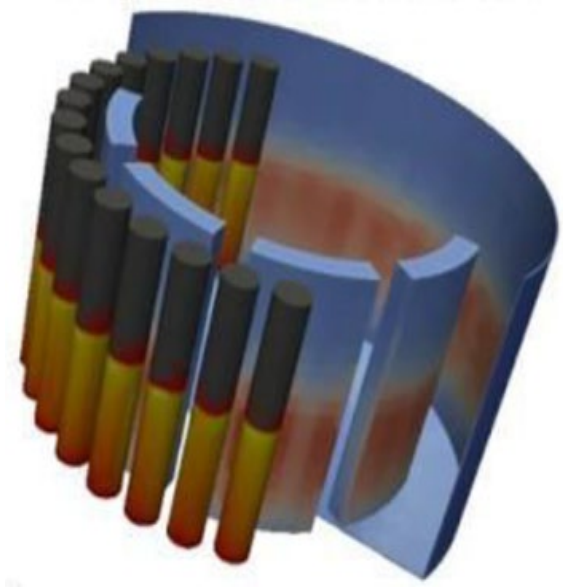


Helium sparging

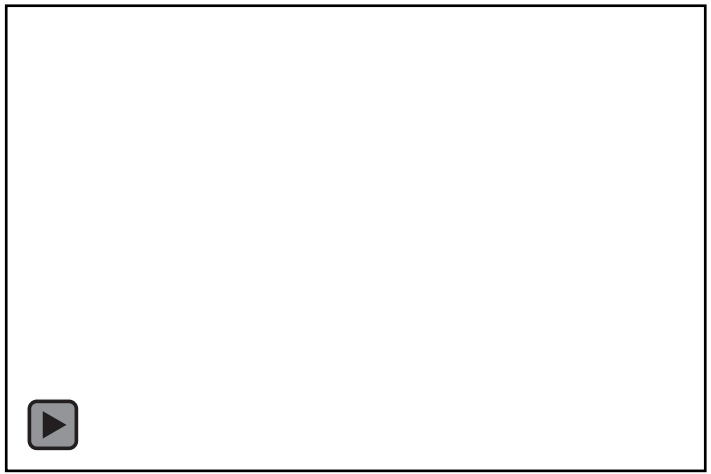


Dump tanks

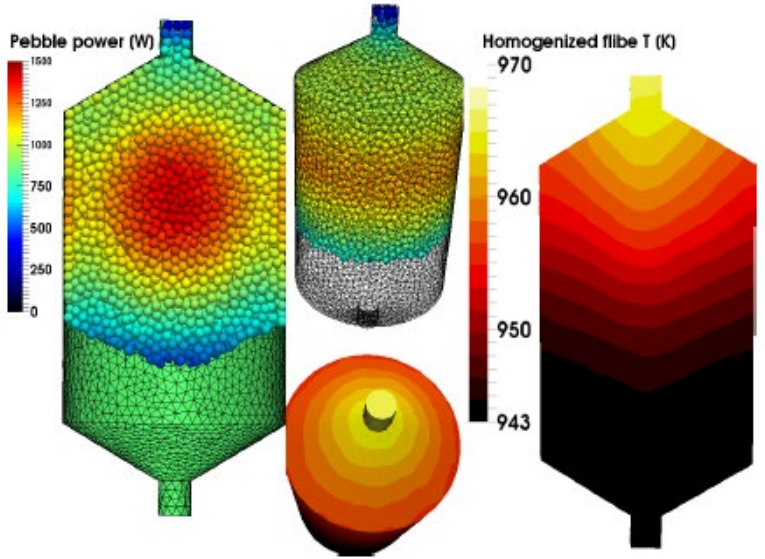
OpenFOAM -> Available two-phase CFD solvers, boundary conditions, radiative heat transfer

FHR modelling (UCB)

- Discrete Element Modeling + coarse-mesh thermal-hydraulics + Monte Carlo



Helium sparging



Dump tanks

OpenFOAM -> Available DEM solver, available multi-physics interface with Serpent, sound solvers (and discretization) for thermal-hydraulics

GeN-Foam: Generalized Nuclear Field operation and manipulation

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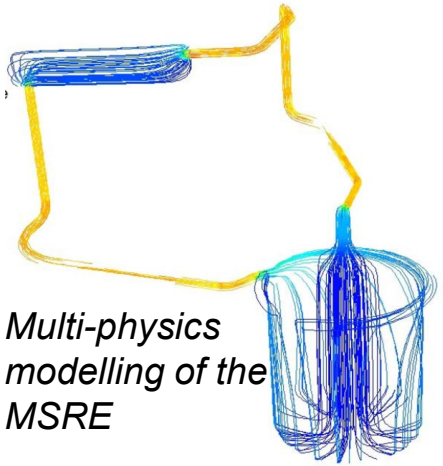
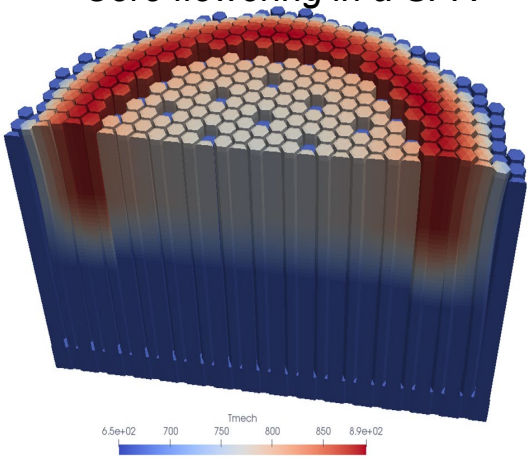
- First general solver for reactor safety based on OpenFOAM

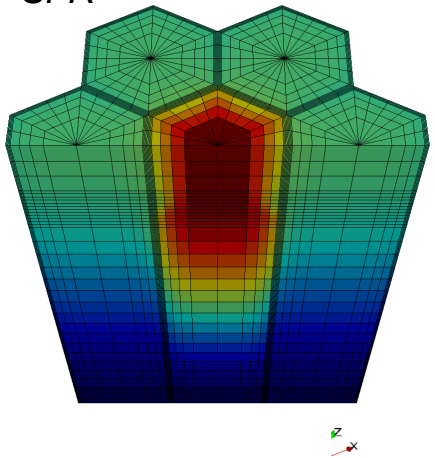


Multi-physics modelling of the MSRE

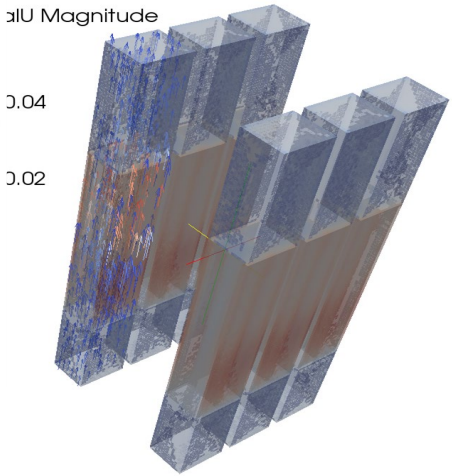
Core flowering in a SFR



Assembly windows in a SFR



The Argonaut reactor

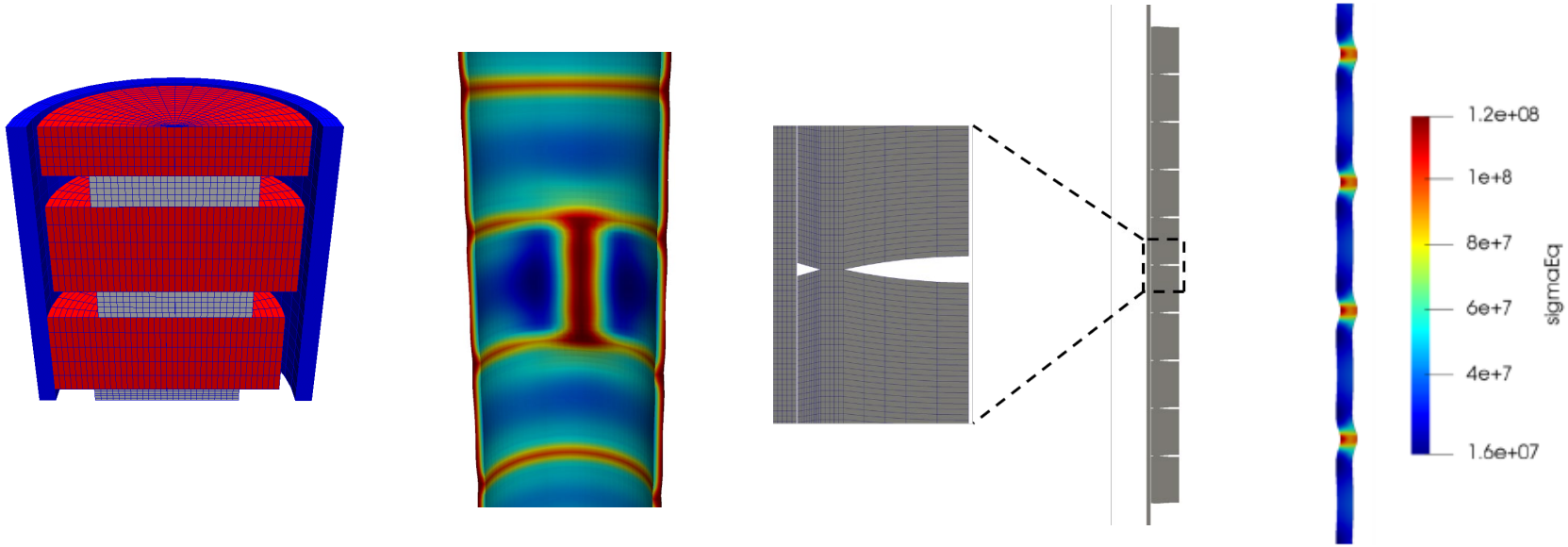


OpenFOAM -> Open-source + object oriented for making use of previous work, multi-mesh with projection algorithms, multi-material, …

OFFBEAT: OpenFoam Fuel BEhavior Tool

Carlo Fiorina

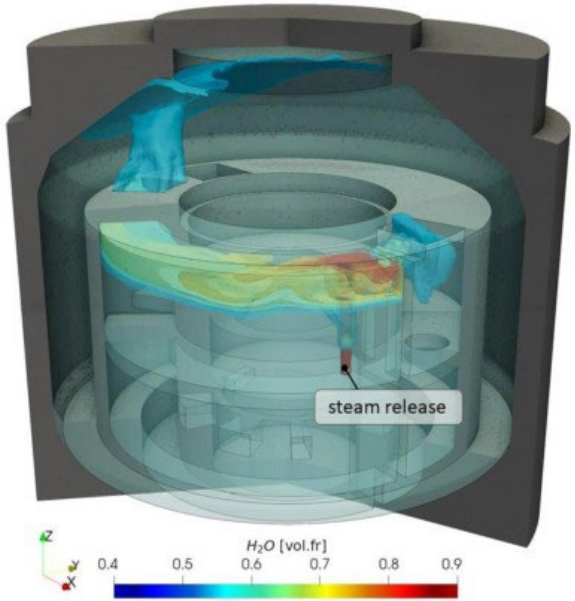
- Thermal-mechanics with finite volumes….



OpenFOAM -> community, intuitive formulation, conservative formulation, available functionalities (region-couple boundaries, …), object-oriented library

HPC-oriented containment analysis

- High-fidelity solver for containment analysis (openly available!)



ISP-37 VANAM-M3 experiment
with containmentFOAM

OpenFOAM -> conservative formulation, available solvers (incl. Monte Carlo!), turbulent models, and available functionalities (region-couple boundaries, …)

One can model pretty much everything…

Lessons learnt

What's the effort?

What competences do I need?

What about the license?

What is the quality of the result?

Downsides

- No graphical user interface (distributed with the code)
- Meshing and post-processing are performed with separate tools
- Meshing often requires proprietary tools
- Requires familiarity with Linux
- Limited documentation

Advantages

- Transparent
- Access to source code

Better integrations of application and development

Structure of the base library

Very complete

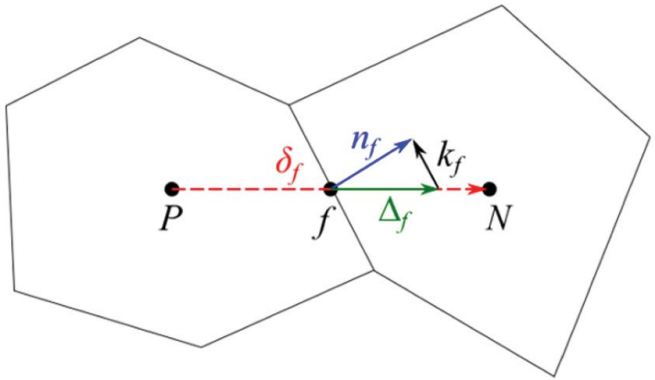
- discretization and linear system solution
- mesh-to-mesh projections
- mesh deformation
- mesh manipulation
- ordinary differential equations
- Monte Carlo methods
- octree-based mesh search
- methods for reduced-order modelling
- built-in and third-party code coupling schemes
- …

Object oriented

- encapsulation
- multi-level API

Finite volumes

- Intuitive
- Conservative
- Optimal for thermal-hydraulics
- Good for thermal-mechanics
- Ok for neutronics

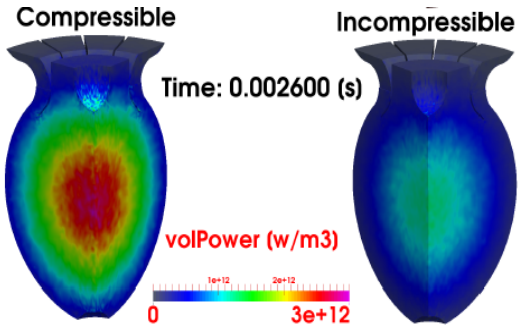


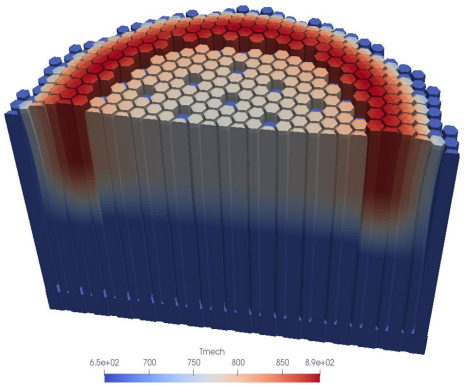
but

- Max second order
- Sensitive to mesh quality

Unstructured meshes

- Complete flexibility in terms of geometry -> non-traditional reactor designs and complex component
- Significant computational footprint
- First order, with all cell faces that are flat -> a high mesh resolution for curved surfaces





Operator-splitting

One matrix for each equation + iteration

Pros

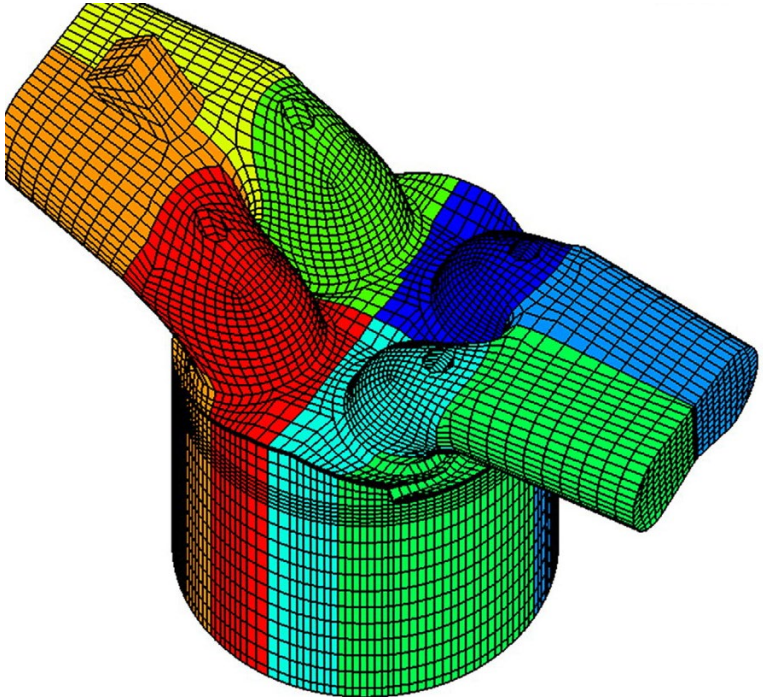
- Easier preconditioning and optimal choice of solution method
- No need to solve all physics at each coupling/time step

Cons

- Can be hard to converge for weakly-coupled / strongly non-linear equations

Parallelization

- Domain decomposition and the MPI
- Optimally scale up to few thousands of CPU cores
- Some bottlenecks
  - ✓ the sub-optimal sparse matrices storage format (LDU) that does not enable any cache-blocking mechanism (SIMD, vectorization)
  - ✓ the I/O data storage system
- The OpenFOAM HPC Technical Committee is currently working on the limitations
  - ✓ interface to external linear algebra libraries
  - ✓ recent work from NVIDIA



Computational requirements

CPU cores

- rule of thumb: 30’000 mesh cells per CPU core
- CFD
  - 2D RANS-> several hundred thousand cells -> 10 CPU cores
  - 3D RANS -> several hundred millions cells -> 5000 CPU cores
- coarse-mesh thermal-hydraulics and neutron diffusion
  - full-core models -> few hundred thousand to few million cells -> workstations or laptops

Runtime

- Steady-state simulations on the optimal number of CPU cores: several minutes to several hours
- Long-running time-dependent problems: up to a week
- In some specific applications, such as detailed containment simulations: up to a month

Memory requirements

- Single-phase RANS CFD simulation -> order of 10 fields -> 1 GB of memory per million cells
- 3D discrete ordinates neutron transport -> several thousand solution fields -> 200 GB of memory per million cells

License

- GNU GPLv3 license
  - copyleft type license: automatically affect derivative work
  - favors a collaborative development with minimal work duplication
  - limits investments from commercial players

Take away messages

- With some practice and commitment OpenFOAM allows one to model pretty much everything, ending up with a solver featuring state-of-the-art numerics
- No need to start from scratch:
  - Many developers will share their work
  - Some solvers are already available online
- Not necessarily an easy tool: Make sure you or your students/employees are relatively familiar with CFD methods, or with numerical methods for PDEs (possibly finite volumes)
- Don’t get frustrated: there is always a way out with OpenFOAM and, most likely, someone who had your same problem and will be happy to help
- Don’t get discouraged: the entry barrier may seem steep, but skills one learns will allow them to tackle many problems
- If possible, do not do it alone!

Thank you

Carlo Fiorina

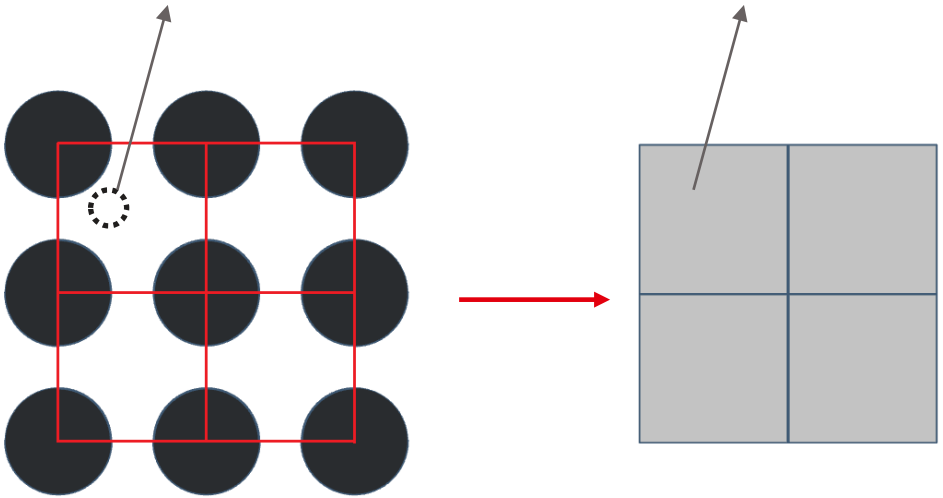
Porous-medium thermal-hydraulics: volume averaging

Carlo Fiorina

$$\langle \phi_i^* \rangle_I = \frac{1}{V_i} \int_{\Omega_i} \phi_i^* \, dV$$

$$\langle \phi_i^* \rangle = \frac{1}{V} \int_{\Omega_i} \phi_i^* \, dV$$

$$\frac{\partial}{\partial t} \rho_i^* + \boldsymbol{\nabla} \cdot (\mathbf{u}_i^* \rho_i^*) = 0 \quad \rightarrow \quad \frac{\partial}{\partial t} (\alpha_i \rho_i) + \boldsymbol{\nabla} \cdot (\alpha_i \mathbf{u}_i \rho_i) = \sum_{j \neq i} \frac{1}{V} \int_{\partial \Omega_{ij}} \rho_i^* (\mathbf{u}_\partial^* - \mathbf{u}_i^*) \cdot \mathbf{n} \, dS$$



Volume averaging results in:

- additional **variables** (phase fraction, tortuosity, etc.);
- additional terms that require **experimentally-informed closure**;

Porous-medium thermal-hydraulics: Governing equations

Carlo Fiorina

The coarse-mesh governing equations (Navier-Stokes and enthalpy) are:

$$\frac{\partial}{\partial t}\left(\alpha_i \rho_i\right) + \boldsymbol{\nabla} \cdot \left(\alpha_i \mathbf{u}_i \rho_i\right) = -\Gamma_{i \to j}$$

$$\frac{\partial}{\partial t}\left(\alpha_i \rho_i \mathbf{u}_i\right) + \boldsymbol{\nabla} \cdot \left(\alpha_i \rho_i \mathbf{u}_i \otimes \mathbf{u}_i\right) =$$

$$- \alpha_i \boldsymbol{\nabla} p + \boldsymbol{\nabla} \cdot \left(\alpha_i \boldsymbol{\sigma}_{d,i}\right) + \alpha_i \rho_i \mathbf{g} - \mathbf{S}_{\mathbf{u},i \to j}$$

$$\frac{\partial}{\partial t}\left(\alpha_i \rho_i h_i\right) + \boldsymbol{\nabla} \cdot \left(\alpha_i \mathbf{u}_i \rho_i h_i\right) =$$

$$\boldsymbol{\nabla} \cdot \left(\alpha_i \kappa_i \boldsymbol{T}_i \cdot \boldsymbol{\nabla} T_i\right) + \alpha_i \frac{\partial}{\partial t} p + \alpha_i \rho_i \mathbf{u}_i \cdot \mathbf{g} + \alpha_i q_{int,i} - S_{h,i \to j}$$

These reduce to traditional CFD approaches in clear fluid regions and a system-code-like approach in 1-D regions (multiple scales).